Externalized Cognitive State Transfer

Compensating for Compositional Consistency Failures in Multi-Step LLM Reasoning via Structured Context Archives

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Abstract

Large language models exhibit compositional consistency failures in multi-step reasoning: when intermediate results from early steps must carry forward to later steps, the models frequently re-derive those results differently, introducing errors that cascade through the remainder of the derivation. We present a protocol-level intervention that compensates for this architectural limitation by externalizing intermediate reasoning state to a structured, hash-chained JSON archive. The protocol — implemented as an append-only, temperature-graded context file — records each reasoning step as an immutable layer with explicit claims, confidence scores, evidence references, and causal links to prior layers. We demonstrate empirically that agents using this protocol achieve correct multi-step derivations across three domains (software engineering, architectural decision-making, and physics) that the same models fail without the intervention. Most notably, we show that a complete cognitive reasoning posture transfers between agent sessions with zero shared context: a new agent with no conversation history loads the archive file and resumes work with full compositional consistency, correctly referencing intermediate results from prior sessions rather than re-deriving them. We term this capability Externalized Cognitive State Transfer and present evidence from cross-model validation (Anthropic Claude and OpenAI Codex) confirming model-agnostic portability of the reasoning archive.

1. Introduction

The emergence of frontier large language models (LLMs) with strong performance on mathematical reasoning benchmarks has led to their increasing deployment in multi-step analytical tasks. However, a growing body of evidence demonstrates that LLM reasoning capabilities degrade significantly as derivation complexity increases, particularly when intermediate results from early reasoning steps must be carried forward and correctly applied in later steps.

Chen et al. (2023) identify two critical failures: *hypothetical consistency* (a model's ability to produce the same answer under semantically equivalent prompt transformations) and *compositional consistency* (consistency of final outputs when intermediate sub-steps are replaced with the model's own outputs for those steps). Even GPT-4 exhibits consistency rates below 50–65% on these measures. Boye et al. (2025) demonstrate that state-of-the-art models produce correct final answers via flawed intermediate logic, exhibit sign propagation errors, and fail to maintain conventions across multi-step derivations.

These failures are not merely quality issues amenable to better prompting or larger context windows. They reflect an architectural limitation: transformer-based models lack persistent intermediate state. Each token is generated from a probability distribution conditioned on the full context window, and as that window grows, relevant intermediate results are diluted by subsequent tokens — a phenomenon termed context rot. The model does not "remember" that it defined $\omega = qB_0/m$ in step 2; it re-infers that definition from the growing context, and each re-inference has a non-trivial probability of introducing variation.

We propose a protocol-level intervention: externalize the intermediate reasoning state to a structured file on disk, where each step is recorded as an immutable, hash-chained layer. When a subsequent step needs an intermediate result, the agent references the layer rather than re-deriving from deteriorating context. We call the resulting capability Externalized Cognitive State Transfer (ECST) and demonstrate that it compensates for compositional consistency failures across three reasoning domains.

2. Related Work

2.1 LLM Reasoning Failures

Song et al. (2026) present a comprehensive taxonomy distinguishing embodied from non-embodied reasoning failures. They identify that LLMs are proficient on shallow one-hop tasks but degrade sharply on multi-hop or compositional problems, an effect traceable to insufficient persistence and propagation of intermediate entities in hidden states. Boye et al. (2025) evaluate eight state-of-the-art models on 50 newly constructed high-school-level word problems and find that all models exhibit errors in spatial reasoning, strategic planning, and arithmetic, sometimes producing correct answers via flawed logic.

The PHYSICS benchmark (ICLR 2025 Workshop) constructed from PhD-qualifying physics exams demonstrates that models struggle with problems requiring multi-step deduction across classical mechanics and quantum physics.

2.2 Context Window Limitations

The NoLiMa benchmark demonstrates that even modern reasoning models degrade as distractor context grows. Research on tool chaining in production LLM agents shows that context preservation across sequential operations is fragile: constraints established early in a chain are frequently lost by later steps, producing outputs that are technically valid but factually wrong. Clemm (2025) identifies context rot as a structural failure mode where accuracy degrades as context grows and the common fix is to start a new chat.

2.3 Existing Approaches

Chain-of-Thought prompting and its extensions (Tree-of-Thought, Self-Consistency sampling) improve performance on individual reasoning steps but do not address inter-step consistency. Process-supervised verification (STEPCO) can detect step errors but requires external verifiers and does not provide persistent state across sessions. RAG stores facts but not reasoning process. Memory systems like Mem0 store preferences but not derivation state. Execution checkpointing (LangGraph) captures program state but not reasoning state. No existing approach provides a structured, verifiable, model-agnostic format for externalizing the complete intermediate reasoning state of a multi-step derivation such that it can be loaded by a different agent in a different session and used to continue the derivation with compositional consistency.

3. Protocol Design

3.1 Core Architecture

The protocol represents a reasoning session as a JSON file containing an immutable core (the stated objective and constraints), a sequence of append-only layers (each representing a reasoning event), and metadata for temperature management, branching, and integrity verification. Each layer contains: a kind (observation, interpretation, decision, synthesis, tool_result); a phase (audit, design, refactor, verification, handoff); a state object with summary, claims, confidence score, and open questions; evidence references linking to prior layers or external artifacts; and a hashes object with the layer hash (SHA-256 of JCS-canonical state per RFC 8785) and chain hash derived from the prior chain hash and current layer hash.

3.2 Temperature-Graded Memory

Layers are organized into three temperature bands: hot (active working set, typically 3–8 layers), warm (recent context, accessible without decompression), and cold (archived history, stored as compressed capsules with zlib+base64 encoding). This geological model solves the fundamental problem of long-running agent context: maintaining relevant working memory without losing the audit trail. The load-context command with a token budget reconstructs the most relevant context within the agent's capacity.

3.3 Branching, Merge, and Integrity

When an agent explores alternative approaches, each path gets its own branch timeline. The protocol enforces single-active-branch governance. Merge operations produce synthesis layers recording which branches were accepted and which were rejected. The core object is hash-guarded at initialization; the engine verifies immutability before every write. Each layer's chain hash is derived via SHA256(prior_chain_hash | layer_hash), creating a walkable chain from genesis to the current head. Any modification after creation is detectable by an independent verifier.

4. Experimental Design

4.1 Models and Configuration

Experiments were conducted using Anthropic Claude (Opus 4.6, via Claude Code) and OpenAI Codex (via CLI). The protocol implementation is a single Python file (771 lines, stdlib only) with zero external dependencies. No model fine-tuning or specialized prompting was used beyond instructing the agent to use the protocol for each reasoning step.

4.2 Domain 1: Software Engineering

A multi-module Python codebase with approximately 30 intentional bugs across four packages (payments, inventory, orders, API) was used as the test subject. The task required security audit, architectural decision-making, implementation, and verification across multiple sessions and agent identities.

4.3 Domain 2: Architectural Decision-Making

A specific decision — choosing between Python Decimal and integer-cents representations for monetary values — required analysis across multiple dimensions, counter-interpretations challenging each approach, and merge with a documented accept/reject decision.

4.4 Domain 3: Multi-Step Physics Derivation

A six-stage derivation of charged particle motion in crossed electric and magnetic fields was selected because it exhibits the documented LLM failure modes. The derivation requires: (1) setting up coupled ODEs from the Lorentz force; (2) defining the cyclotron frequency and decoupling; (3) solving via two independent methods with branching; (4) applying initial conditions using prior intermediate results; (5) integrating velocity to position with differentiation verification; (6) deriving drift velocity, checking dimensions, and verifying limiting cases. An automated verification script checks the archive against 22 specific intermediate results from the known-correct solution.

5. Results

5.1 Software Engineering: Cross-Session Transfer

The task was conducted across four autonomous sessions using Claude Code, with subsequent sessions using OpenAI Codex for cross-model validation.

Session	Agent	Layers	Phases	Depth	Evidence	Loops
1: Security audit	Claude	6	audit	5	6 refs	3
2: Branch refactor	Claude	13	audit, design	12	18 refs	2
3: Bug fix (v0.2)	Claude	6	3 phases	5	6 refs	1
4: Full cleanup	Codex	19	6 phases	18	65 refs	1
5: Blank resume	Claude	+layers	verification	17	+refs	0
6: Test expansion	Claude	+layers	verification	17	65+	0

Critical result: Session 5 began with zero conversation history. The agent received only “Load PEARL and resume.” It loaded the archive, reconstructed full cognitive state, identified a real contradiction between the archive and the codebase, corrected it, verified 13 tests, verified the hash chain, and documented its own arrival as a new layer. A subsequent session expanded coverage to 34 tests and resolved the last deferred item from a prior agent’s decision. Final state: 4 sessions, 3 agent identities, depth 17, 34 tests, zero open loops.

5.2 Architectural Decision-Making: Branching and Deliberation

The monetary representation decision was recorded with two explicit branches. Counter-interpretations challenged both approaches: against cents-integer, that round() has banker’s rounding surprises; against Decimal, that the Decimal(float) constructor trap is more dangerous than the original bug. The synthesis records: cents-integer accepted, decimal-approach rejected, with a hybrid recommendation. The branch metadata is permanently auditable.

5.3 Physics Derivation: Compositional Consistency Preserved

The six-stage derivation was executed with the protocol tracking every intermediate result. The archive contains 9 layers across 4 phases with 17 evidence references creating an explicit dependency chain:

Stage	Kind	Key Result	Claims	Evidence
1	observation	Coupled ODEs from Lorentz force	3	problem statement
2	interpretation	$\omega = qB_0/m$, decoupled ODE	4	← Stage 1
3A	interpretation	Complex variable solution	4	← Stage 2
3B	interpretation	Direct ODE solution	3	← Stage 2
3'	synthesis	Both methods agree	4	← 2, 3A, 3B
4	decision	$A = v_0 - E_0/B_0$, $B = 0$	4	← Stages 2, 3'
5	tool_result	Cycloid equations	4	← Stages 2, 4
6	interpretation	Drift, limits verified	5	← Stages 2, 5

Verification result: 22/22 automated checks passed against the known-correct solution. Every intermediate result — cyclotron frequency, sign conventions, particular solution, initial condition coefficients, cycloid form, drift velocity, and both limiting cases — was verified correct.

5.4 Cross-Session Physics Extension

In a subsequent session, the agent loaded the completed archive and extended the result to a magnetic mirror field $B = B_0(1 + \alpha z)\hat{z}$. The agent's first action was to create an observation layer explicitly importing $v_d = E_0/B_0$ from the archived Stage 6 layer by layer ID, with the claim: "This extension uses the PEARL-recorded drift velocity directly as prior state. No re-derivation of the E×B drift is performed." The agent correctly derived that the gradient-B drift vanishes for this configuration ($B \times \nabla B = 0$), distinguished it from the E×B drift, and added a caveat about $\nabla \cdot B = 0$ compliance. The hash chain verified intact after extension.

5.5 Cross-Model Validation

The software engineering experiments were conducted on both Anthropic Claude Code and OpenAI Codex, with both agents operating on the same archive files. Both models successfully loaded, interpreted, and extended archives created by the other. This confirms model-agnostic portability in practice.

6. Analysis

6.1 Externalized vs. Internalized State

The protocol converts the problem from inference to reference. Without it, an agent reaching Stage 5 must infer ω from growing context — scanning backward through pages of algebra. Each such inference has a probability p of introducing variation. Over N inferences, the probability of

at least one error is $1 - (1-p)^N$. With the protocol, the agent reads ω from the specific layer where it was defined, identified by layer ID. The probability of error drops to effectively zero.

6.2 Temperature and Relevance

The temperature model addresses context window limitations directly. Hot layers contain the current working set; warm contains recent context; cold contains compressed history. This is functionally equivalent to how human experts maintain working memory: the theorem you proved last week is available if needed, but you don't re-derive it every time you use it.

6.3 Limitations

The protocol requires agent cooperation — the quality of the archive depends on the quality of layers the agent creates. The protocol does not verify correctness of claims; it verifies only integrity. The physics verification required a separate automated checker. Cross-model experiments were conducted on specific tasks; generalization to other domains remains to be validated, though structural parallels are strong.

7. Implications

If compositional consistency failures are architectural, and if externalizing state compensates for them, then the protocol represents a new layer in the agent architecture stack: a cognitive persistence layer sitting between the model (reasoning capability) and the runtime (tool access), providing structured memory that neither currently offers. The hash-chained, append-only structure provides a tamper-evident reasoning audit trail mapping to NIST SP 800-53 controls. The physics derivation demonstrates potential as a structured lab notebook for AI-assisted scientific work, where a reviewer can inspect not just the final result but the complete reasoning process including failures and alternatives.

8. Conclusion

We have presented Externalized Cognitive State Transfer — a protocol-level intervention compensating for compositional consistency failures in multi-step LLM reasoning by externalizing intermediate state to a structured, hash-chained JSON archive. Validated across three domains:

1. **Software engineering:** 4 sessions, 3 agent identities, cross-session transfer with zero shared context, depth 17, 34 tests, zero open loops.
2. **Architectural decision-making:** Branching with counter-interpretations, accept/reject merge, auditable synthesis.
3. **Physics derivation:** 22/22 automated checks on a 6-stage electromagnetic derivation. Zero re-derivation errors, zero sign failures, zero lost terms.

Cross-model validation confirms model-agnostic portability. Cross-session transfer demonstrates cognitive state persistence independent of the agent, platform, and conversation history.

The agent is ephemeral. The reasoning persists.

References

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Appendix A: Empirical Archive Summary

Archive	Version	Layers	Branches	Phases	Ev. Refs
charged-particle.pearl	0.1	8	3 (A, B, main)	derivation	implicit
charged-particle-v02.pearl	0.2	9	1 (main)	4 phases	17
session-pre-cleanup.pearl	0.2	6	1 (main)	3 phases	6
session.pearl (full)	0.2	19	3 (main, dec, int)	6 phases	65

All archive files, the automated physics verification script, and the test projects are available as supplementary materials.

Appendix B: Protocol Specification

File extension: .pearl

Media type: application/vnd.pearl.chat+json

Encoding: UTF-8 JSON, JCS-RFC 8785 canonical serialization

Hash algorithm: SHA-256 (FIPS 140-3 compatible)

Chain derivation: SHA256(prior_chain_hash | layer_hash)

Core guard: _core_hash verified before every write

Implementation: Python 3.10+, 771 lines, zero external dependencies

Validated models: Anthropic Claude (Opus 4.6), OpenAI Codex